Does L+S Beat Either Band Alone?

Surface-water mapping and change detection with NISAR L- and S-band GCOV over the Bihar Ganga floodplain, monsoon 2026

Team Drishya

InGARSS 2026 Young Professionals Hackathon, Phase 1

Problem and Objective

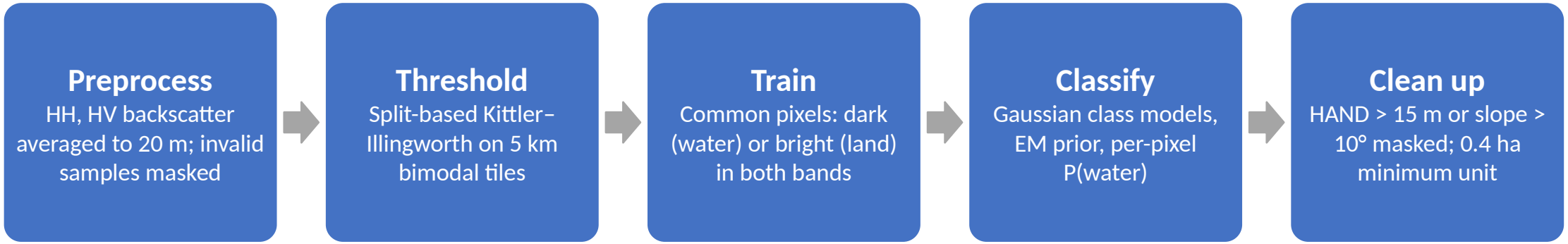
- Map surface water and its change (stable water, new water, water loss) over the full NISAR frame around Bihar's Ganga floodplain (25.5° N, 86.5° E) during the 2026 monsoon
- Do it three ways, using only NISAR L2 GCOV data: L-band only, S-band only and L+S combined
- Core question: does combining L- and S-band measurably beat either band alone?
- Fair comparison: one algorithm, one set of training pixels and one post-processing chain; only the input features change
- Every claim validated against independent Sentinel-2 data, with confidence intervals

Study Area and Data

	L-band (NASA / ASF)	S-band (ISRO / Bhoonidhi)
Product	L2 GCOV, PROVISIONAL (calibrated)	L2 GCOV, PROVISIONAL (12 Sep: urgent-response only)
Mode and polarizations	40+5 MHz, HH/HV, 10 m	37 MHz, HH/HV, 10 m
Joint L+S dates (UTC)	14 Jul, 12 Sep, 18 Oct 2026	14 Jul, 12 Sep, 18 Oct 2026 (none before 14 Jul)
Looks per 20 m pixel	about 19	about 64
Polarimetry pass	14 Sep descending, quad-pol	14 Sep descending, compact-pol

- Frame: ascending track 98, frame 14, about 250 × 250 km; common 20 m grid in UTM 45N (EPSG:32645)
- Supporting layers: Copernicus GLO-30 HAND and slope, ESA WorldCover 2021, JRC Global Surface Water
- Reference: Sentinel-2 L2A MNDWI on 18 Oct (same UTC day) and 13 Sep (5.4 h after the peak pass)

Methodology: Water Mapping



- Feature sets: L = [L-HH, L-HV]; S = [S-HH, S-HV]; L+S = all four channels (Bayesian likelihood fusion)
- Identical algorithm, settings, training pixels, clean-up and validation pixels for every feature set
- References: naive L AND S and L OR S; equal-information control = L-band at 40 m (about 4× the looks)
- Uncertain zone: $0.2 < P(\text{water}) < 0.8$

Change Detection and Validation

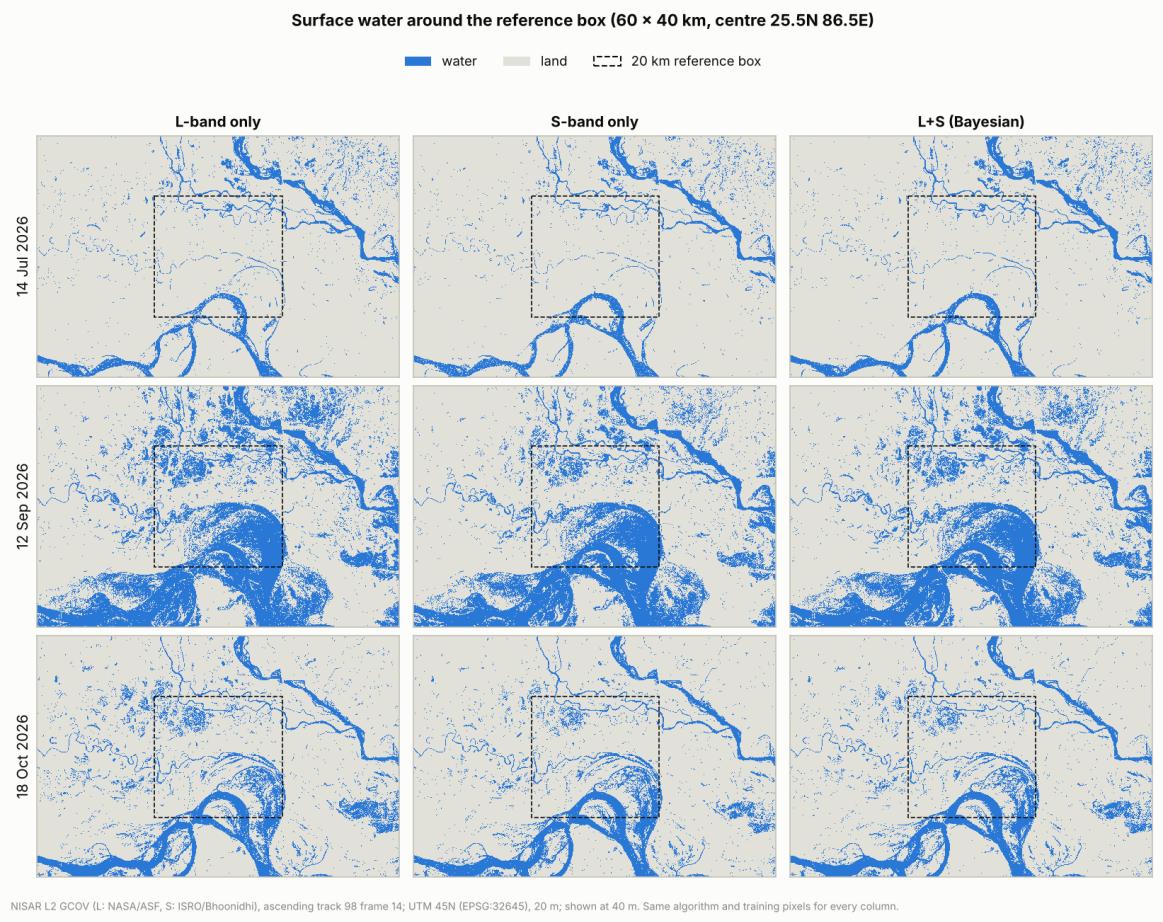
Change detection

- Post-classification comparison of the per-date maps
- Classes: stable water, new water, water loss, stable land
- Per-pixel confidence: P(gain) = P2 × (1 − P1)
- New water outside JRC persistent water (≥ 75%) = potential temporary surface inundation
- Gain: 14 Jul → 12 Sep; loss: 12 Sep → 18 Oct

Validation

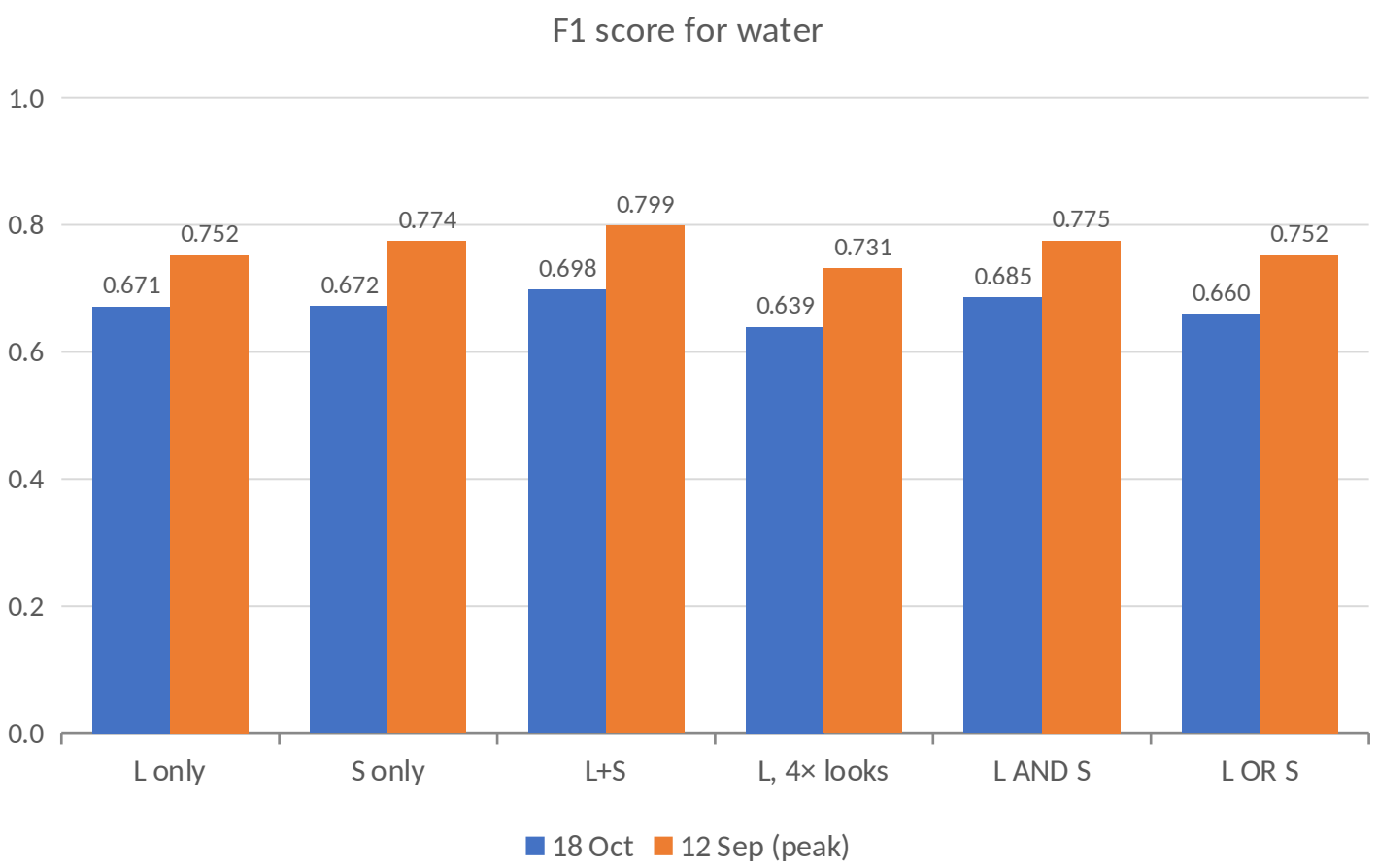
- Reference: Sentinel-2 L2A MNDWI > 0, clouds masked with SCL
- 18 Oct: 36,400 km² clear; 13 Sep: 21,900 km² clear
- Every map scored on identical pixels
- F1, precision, recall, IoU; 95% CIs from a 5 km block bootstrap
- McNemar tests; error-adjusted areas (Olofsson et al. 2014)

Results: Surface-Water Maps



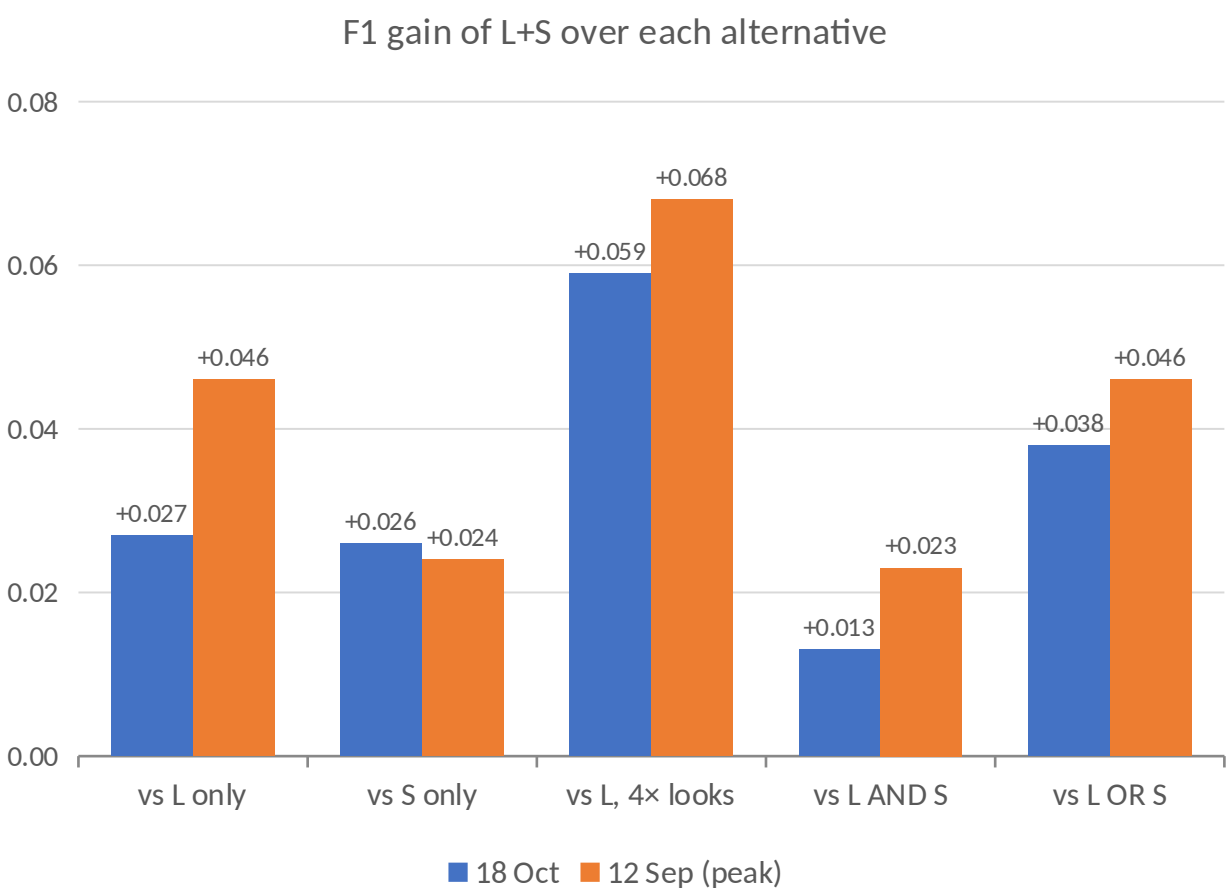
- 14 Jul: channels near low water; 2,866 km² mapped with L+S
- 12 Sep: the Kosi–Ganga floodplain fills; 5,630 km²
- 18 Oct: receding; L 2,759, S 2,374, L+S 2,464 km²
- L-band maps the most water, S-band the least
- Error-adjusted areas agree across methods: about 6,000 km² at the peak, 2,700 km² on 18 Oct

Results: Accuracy against Sentinel-2



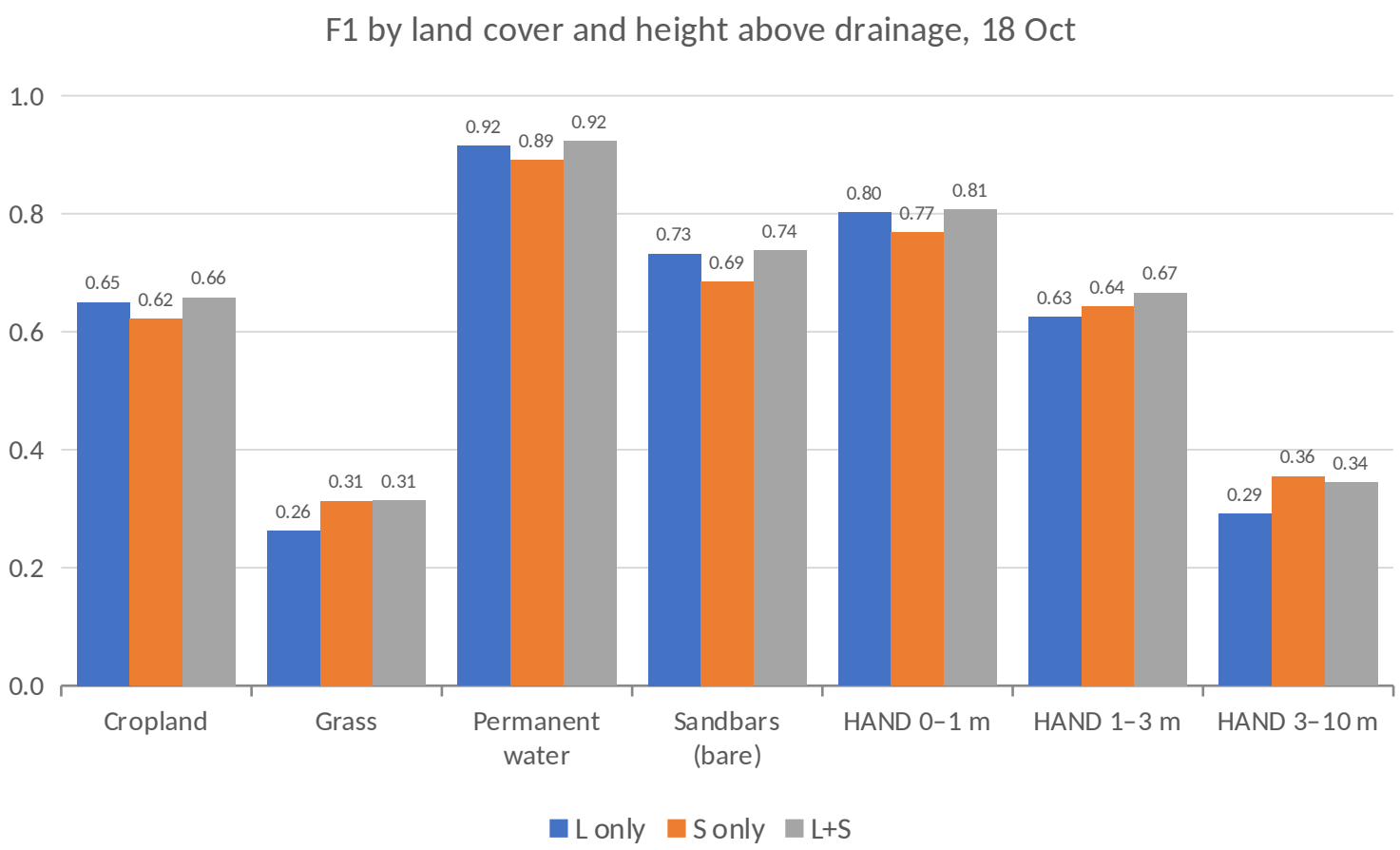
- L+S is the most accurate of six maps on both dates
- 18 Oct: F1 0.698 (95% CI 0.674–0.719) vs 0.671 (L) and 0.672 (S)
- 12 Sep: F1 0.799 (0.777–0.817) vs 0.752 (L) and 0.774 (S)
- L-band with 4× the looks scores lowest

Added Value of Combining L- and S-band



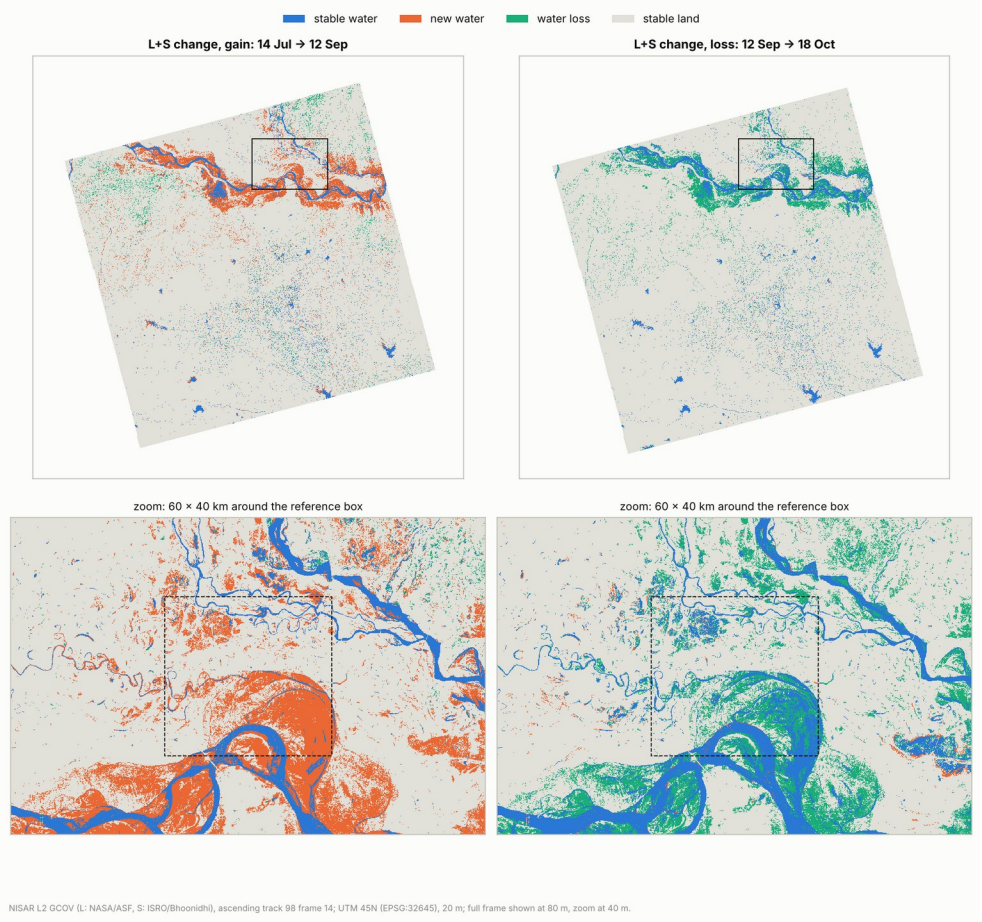
- Gains over each band have 95% CIs that exclude zero: +0.027 [+0.022, +0.033] and +0.046 [+0.041, +0.052] vs L; +0.026 and +0.024 vs S
- Complementary errors: L finds more water but also more false water; S is more precise but misses water; fusion keeps both strengths
- Not just more data: L-band with 4× the looks is worse
- Band choice changes the flood estimate: new water 4,501 km² (L) vs 3,616 km² (S)

Where Does Fusion Help?

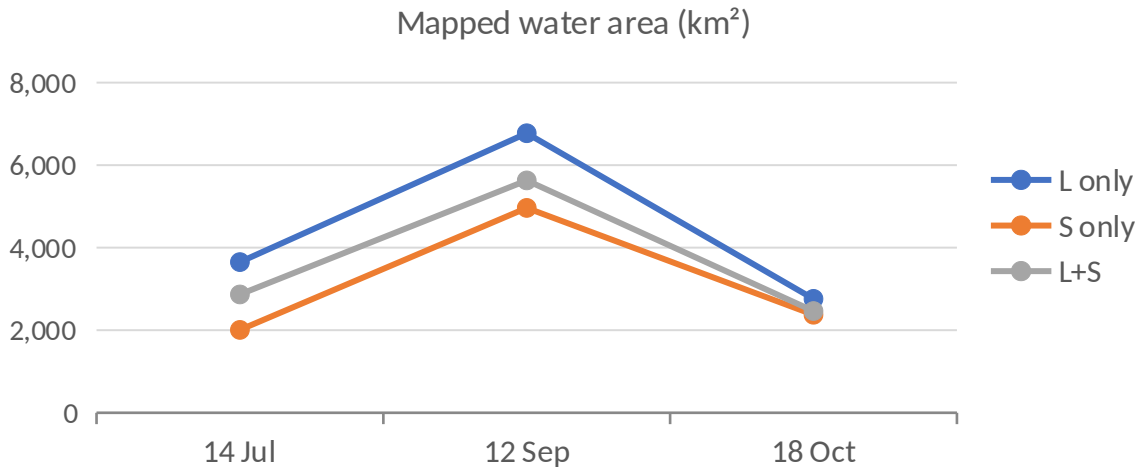


- Largest gain just above the floodplain: HAND 1–3 m, F1 0.666 vs 0.625 (L) and 0.643 (S)
- L leads on cropland and sandbars, so “S-band fixes sandbars” is not supported
- S leads on grass and wetland
- L+S matches or beats the better band in nearly every class

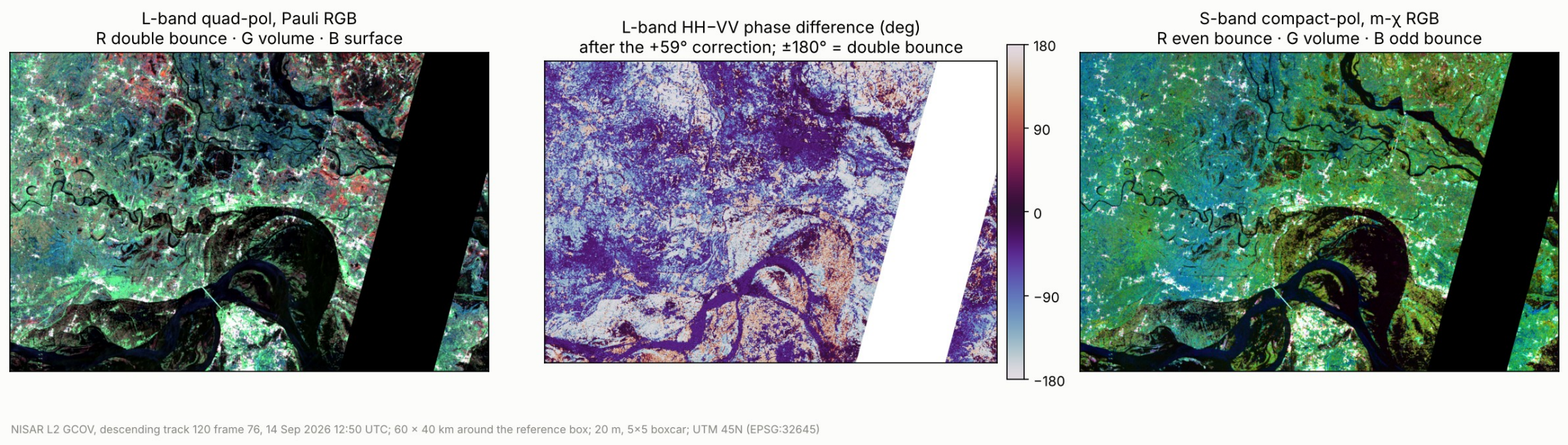
Results: Gain and Recession (L+S)



- Gain, 14 Jul → 12 Sep: 3,774 km² of new water, 3,758 km² of it outside persistent water
- 80% on cropland (3,038 km²); 79% within 1 m of the nearest drainage
- Loss, 12 Sep → 18 Oct: 3,273 km² receded; 2,266 km² stayed water



Polarimetry (14 Sep, Descending)



- L-band quad-pol: Pauli decomposition and HH–VV phase difference
- 59° phase calibration correction, sign verified: open water −20°, double bounce +167°
- 6.1% of the window shows double bounce: candidate flooded vegetation
- S-band compact-pol m-χ separates rivers, vegetation and settlements

Limitations and Future Work

Limitations

- Optical reference misses water under vegetation; the 18 Oct image is 19 h before the pass; 13 Sep is only 31% clear
- S-band only from 14 Jul; 12 Sep S-band is urgent-response; ISRO's sample-shift flag covers 36% of the S swath
- 20 Jun (L-band only) excluded: dry pre-monsoon fields look like water
- No banding correction; one frame and track

Future work

- Explain the 0.4 dB S–L offset inside the sample-shift flag, by land cover
- Incidence-angle banding correction; more dates (31 Aug, 6 Oct)
- Labelled sample points for cloudy dates and flooded vegetation
- Polarimetric features inside the classifier

Conclusions

- Yes: combining L- and S-band measurably improves NISAR surface-water mapping
- F1 gains of +0.024 to +0.046 over the single bands, significant on both validated dates
- The gain comes from the second frequency, not from extra looks
- 2026 monsoon: 3,774 km² of new water between 14 Jul and 12 Sep, mostly cropland in the low floodplain; 3,273 km² receded by 18 Oct
- Fully reproducible: github.com/Vamsi12345-eng/INGARSS (one command per step)

Thank you! Team Drishya